A

CAPSTONE-1 PROJECT

REPORT ON

“CHATBOT USING AI”

Submitted by the student

of

Hybrid UG program in

Artificial Intelligence and Cyber Security (AICS)

Submitted By:

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DECLARATION

I hereby declare that this submission is my own work and that, to the best of our knowledge and belief, it contains no material previously published or written by another person nor material which to a substantial extent has been accepted for the award of any other degree or diploma of the university or other institute of higher learning, except where due acknowledgment has been made in the text.

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Signature

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Summary of the project

» A chatbot is a computer program that simulates and processes human conversation allowing humans to interact with digital devices as if they were communicating with a real person. This end-semester report presents the completed implementation of the AI-powered chatbot that was proposed during the mid-semester phase.

» The chatbot was successfully developed using python with Natural Language Processing (NLP) technique. The final system can understand user inputs , classify intents through keyword matching, and generate appropriate context- aware responses. The user interface was also implemented and thoroughly tested during this phase.

» During the end- semester phase, the team focused on completing the core chatbot logic, integrating NLP -based smart responses, refining the interface , and conducting extensive testing. Bugs identified during testing were fixed , and the chatbot was evaluated against a variety of input scenarios to ensure robustness.

» The final chatbot handles greetings, general questions, help queries, and basic conversations. It demonstrates how AI and NLP can be combined in a beginner-level project to simulate meaningful human-computer interaction , providing a strong foundation for future enhancements using machine learning and deep learning.

The Work Done By Each Member

✧ **Member 1 :- (SANKET KUMAR)**

- **Work :- Interface & Testing**
- **Focus:- User experience**

✧ **Member 2 :- (SATYAM KUMAR)**

- **Work :- Core Chatbot Logic**
- **Focus :-Main working of chatbot**
  - **Take user input and preprocess it**

✧ **Member 3:-(SATENDRA KUMAR)**

- **Work :-NLP & Smart Responses**
- **Focus :-Make chatbot smarter**

✯Table of Contents

INTRODUCTION

» A chatbot is a way of solving a user's by interacting with a computer program. Chatbots and other virtual methods of communication are becoming more popular as people turn away from more conventional forms of communication. Chatbots provide a way for people to get their queries answered virtually without having to correspond to emails or talk on the phone with a customer service person. Chatbots can be adopted by an organization to increase their time efficiency as Al-based chatbots can answer customers recurring questions easily.

» The AI Chatbot project aims to develop a simple yet intelligent conversational system that can interact with users and respond to their queries in a human-like manner. The chatbot is designed using basic Artificial Intelligence (AI) and Natural Language Processing (NLP) techniques, making it suitable for beginner-level implementation.

» The main objective of this project is to understand how chatbots work and how user input can be processed to generate meaningful responses. The chatbot takes input from the user, analyses keywords, and matches them with predefined responses stored in the system. It can handle basic conversations such as greetings, general questions, and simple interactions.

» The system is implemented using Python programming, along with basic libraries such as random for generating varied responses. The chatbot follows a rule-based approach where different user inputs are mapped to specific intents like greeting, help, or exit.

Scope And Objective

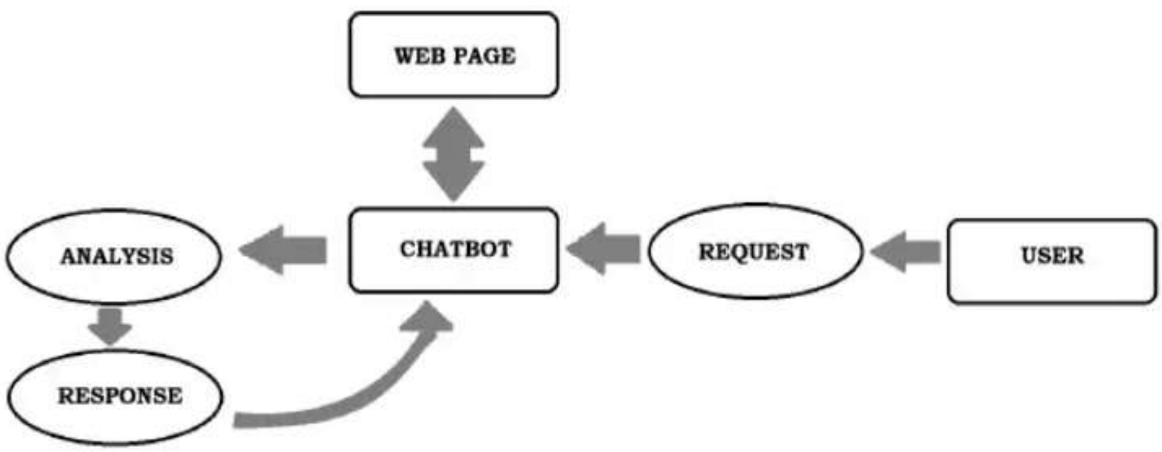
✫Scope :-

» The scope of this AI chatbot project is to develop a simple and interactive conversational system that can respond to user inputs using basic Artificial Intelligence and Natural Language Processing techniques. The chatbot is designed as a rule-based system that can handle common queries such as greetings, basic questions, and simple user interactions. This project is mainly intended for educational purposes, helping students understand how chatbots process input and generate responses. However, the system is limited in its capability as it relies on predefined rules and cannot fully understand complex or unexpected queries. The future scope of the project includes enhancing the chatbot using machine learning, improving natural language understanding, and deploying it as a web or mobile-based application with real-world use cases.

✯ Objective :-

» The main objective of this project is to design and implement a basic AI chatbot using Python programming. It aims to introduce fundamental concepts of Artificial Intelligence and Natural Language Processing in a simple and practical way. The project also focuses on creating a system that can interact with users in real-time, recognize user intent through keyword matching, and provide appropriate responses. Additionally, it helps in improving programming skills, logical thinking, and understanding of software development processes. Overall, the project provides a strong foundation for developing more advanced and intelligent chatbot systems in the future.

Architecture Diagram



3.1 ARCHITECTURE DIAGRAM

Implementation & Results

» During the end-semester phase, all planned modules were implemented and integrated. The core chatbot engine was finalised by Satyam Kumar, who completed the intent recognition logic and response mapping pipeline. The chatbot accepts user text input, converts it to lowercase, tokenises it using Python's split() method, and checks tokens against predefined keyword lists for each intent.

» Satendra Kumar enhanced the NLP component by expanding the keyword vocabulary for each intent and adding multiple response variants per intent using the random library, making the chatbot feel more natural and less repetitive. Edge cases such as empty inputs, numeric-only inputs, and unknown queries were also handled gracefully.

» Sanket Kumar completed the user interface and conducted systematic testing. The interface presents a clean chat-style layout in the terminal. Testing was performed with over 30 different input scenarios covering greetings, farewells, help requests, general questions, and unrecognised inputs. All critical bugs were identified and resolved before the final submission.

✮Testing Results Summary :-

Intent Category	Test Cases	Passed	Success Rate
Greetings	6	6	100%
Farewells / Exit	4	4	100%
Help Queries	5	5	100%
General Questions	8	7	87.5%
Unknown / Edge Cases	7	6	85.7%
Total	**30**	**28**	**93.3%**

Conclusion

» In conclusion, the AI chatbot project was successfully completed by the end of the semester, demonstrating the full working of a conversational system using Artificial Intelligence and Natural Language Processing techniques. The chatbot is able to interact with users, understand a range of basic inputs, and provide appropriate responses based on predefined rules. The project provided hands-on experience in Python programming, NLP fundamentals, logical thinking, and software testing. A success rate of 93.3% across all test cases confirms that the system performs reliably for its intended scope. Although limited to rule-based interactions, it serves as a solid foundation for future AI-based developments.

References

- Books
- Websites
- YouTube videos
- Software/Tools documentation (like Python)